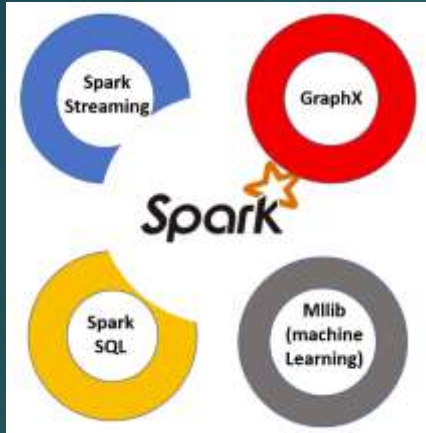




Big Data – Project3

VEDRAN MITIĆ 2123

Step 1

- ▶ 5 files:
docker-compose.yml – for Docker cluster
prepare_data.py – for fetching desired amount of data
train_model.py – for preprocessing and training data
predict.py – for test data and evaluation
visualize.py – for plotting the results
- ▶ Run prepare_data.py:
python prepare_data.py --source ./downloaded_data --target-gb 1.5
- ▶ **--target-gb ~ specify targeted amount of data out of 4.14GB of original data**
- ▶ To get downloaded_data follow **Project1** instructions
- ▶ prepare_data.py – generates the dataset_ready_for_hdfs folder.



Step 2

- ▶ Run: `docker-compose up -d`
- ▶ Images:
 - `bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8`
 - `bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8`
 - `bde2020/spark-master:3.1.1-hadoop3.2`
 - `bde2020/spark-worker:3.1.1-hadoop3.2`

☐	↓	Name	Container ID	Image	Port(s)	CPU (%)	Memory	Actions
☐	▼	● <u>project3</u>	-	-	-	1.25%	1.05GB /	  
☐	●	namenode	ccfe5e346080 	bde2020/h	9000:9000  Show all ports (2)	0.29%	401.4MB	  
☐	●	datanode	f77abf6f139d	bde2020/h		0.47%	282.5MB	  
☐	●	spark-master	9de5d30478c8	bde2020/s	4040:4040  Show all ports (3)	0.13%	206.7MB	  
☐	●	spark-worker	8953b559fe5b	bde2020/s	8081:8081 	0.36%	183.9MB	  

Step 3

- ▶ Create directory and send files to HDFS, run:
docker exec -it namenode hdfs dfs -mkdir -p /user/root
docker exec -it namenode hdfs dfs -put /data/incoming /user/root/dataset

DFS Used:

1.51 GB (0.15%)

/user/root/dataset

Go!

Show

25

▼

entries

Search:

☐	 Permission	 Owner	 Group	 Size	 Last Modified	 Replication	 Block Size	 Name	
☐	 -rw-r--r--	 root	 supergroup	 10.23 MB	 Mar 26 22:25	 1	 128 MB	 uq_vsd_case01_fulldata_07.csv	
☐	 -rw-r--r--	 root	 supergroup	 10.3 MB	 Mar 26 22:25	 1	 128 MB	 uq_vsd_case01_fulldata_08.csv	
☐	 -rw-r--r--	 root	 supergroup	 10.43 MB	 Mar 26 22:25	 1	 128 MB	 uq_vsd_case01_fulldata_10.csv	
☐	 -rw-r--r--	 root	 supergroup	 10.27 MB	 Mar 26 22:25	 1	 128 MB	 uq_vsd_case01_fulldata_11.csv	
☐	 -rw-r--r--	 root	 supergroup	 10.23 MB	 Mar 26 22:25	 1	 128 MB	 uq_vsd_case01_fulldata_12.csv	
☐	 -rw-r--r--	 root	 supergroup	 11.16 MB	 Mar 26 22:25	 1	 128 MB	 uq_vsd_case03_fulldata_05.csv	
☐	 -rw-r--r--	 root	 supergroup	 11.24 MB	 Mar 26 22:25	 1	 128 MB	 uq_vsd_case03_fulldata_11.csv	
☐	 -rw-r--r--	 root	 supergroup	 11.25 MB	 Mar 26 22:25	 1	 128 MB	 uq_vsd_case03_fulldata_12.csv	

Step 4

- ▶ Since bde images are missing numpy which is required for MLlib, you have to install it on spark services. Run:
docker exec spark-master2 apk add --no-cache py3-numpy
docker exec spark-worker apk add --no-cache py3-numpy

Step 5

- ▶ Training, run the command:
`docker exec -it spark-master2 /spark/bin/spark-submit --master spark://spark-master:7077 --driver-memory 2G --executor-memory 4G /app/train_model.py`
- ▶ **You can add `--task "classification"` or `"regression"`**
- ▶ Two ML tasks:
 - Classification: $SpO_2 < 95\%$ → hypoxemia (label=1), else normal (label=0)
 - Regression: predict Heart Rate (HR) from the other vital-sign features
- ▶ Train/validation/test split – 70%/15%/15%
- ▶ ParamGridBuilder – for testing purposes is given only one hyperparameters value for each Grid.
- ▶ 3-Fold Cross-Validation
- ▶ Two ML Algorithms: Random Forest and Gradient-Boosted Trees
- ▶ Metrics:
 - Classification – F1, Accuracy, Precision, Recall
 - Regression – RMSE, MAE, R2

- Workers (1)

Worker Id	Address	State	Cores	Memory	Resources
worker-20260327134647-172.18.0.5-36685	172.18.0.5:36685	ALIVE	4 (4 Used)	8.0 GiB (4.0 GiB Used)	

- Running Applications (1)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
app-20260327140108-0000	(kill) BDS_Project3_VitalSigns	4	4.0 GiB		2026/03/27 14:01:08	root	RUNNING	29 min

Spark Jobs (?)

User: root
Total Uptime: 25 min
Scheduling Mode: FIFO
Active Jobs: 1
Completed Jobs: 354

▶ Event Timeline

▼ Active Jobs (1)

Page: 1

Job Id ▼	Description	Submitted	Duration	Stages: Succeeded/Total
354	aggregate at DecisionTreeMetadata.scala:125 aggregate at DecisionTreeMetadata.scala:125 (kill)	2026/03/27 14:25:59	6 s	0/1

Page: 1

▼ Completed Jobs (354)

Page: 1 2 3 4 >

Job Id ▼	Description	Submitted	Duration	Stages: Succeeded/Total
353	take at DecisionTreeMetadata.scala:119 take at DecisionTreeMetadata.scala:119	2026/03/27 14:25:59	35 ms	1/1

Stages for All Jobs

Active Stages: 1

Pending Stages: 1

Completed Stages: 702

Skipped Stages: 40

▼ Active Stages (1)

Page: 1

Stage Id ▼	Description		Submitted	Duration	Tasks: Succeeded/Total
742	treeAggregate at Statistics.scala:58	+details (kill)	2026/03/27 14:26:57	Unknown	0/12

Page: 1

► Pending Stages (1)

► Completed Stages (702)

► Skipped Stages (40)

Storage

- RDDs

ID	RDD Name	Storage Level	Cached Partitions	Fraction Cached	Size in Memory	Size on Disk
35	*(1) Sample 0.0, 0.7, false, 11 +- *(1) Sort [Time#16 ASC NULLS FIRST, RelativeTimeMilliseconds#2223 ASC NULLS FIRST, Clock#18 ASC NULLS FIRST, HR#2290 ASC NULLS FIRST, ST_II#2357 ASC NULLS FIRST, Pulse#2424 ASC NULLS FIRST, SpO2#2491 ASC NULLS FIRST, Perf#23 ASC NULLS FIRST, etCO2#2625 ASC NULLS FIRST, ImCO2#2692 ASC NULLS FIRST, awRR#2759 ASC NULLS FIRST, NBP_Sys#2826 ASC NULLS FIRST, NBP_Dia#2893 ASC NULLS FIRST, NBP_Mean#2960 ASC NULLS FIRST, NBP_Pulse#3027 ASC NULLS FIRST, NBP_Time_Remaining#3094 ASC NULLS FIRST, ART_Sys#3161 ASC NULLS FIRST, ART_Dia#3228 ASC NULLS FIRST, ART_Mean#3295 ASC NULLS FIRST, etDES#35 ASC NULLS FIRST, inDES#36 ASC NULLS FIRST, etISO#37 ASC NULLS FIRST, inISO#38 ASC NULLS FIRST, etSEV#39 ASC NULLS FIRST, ... 45 more fields], false, 0 +- *(1) Project [Time#16, cast(RelativeTimeMilliseconds#17 as double) AS RelativeTimeMilliseconds#2223, Clock#18, cast(HR#19 as double) AS HR#2290, ST-II#20 AS ST_II#2357, cast(Pulse#21 as double) AS Pulse#2424, cast(SpO2#22 as double) AS SpO...]	Disk Memory Deserialized 1x Replicated	12	100%	0.0 B	4.0 GiB
48	*(1) Sample 0.7, 0.85, false, 11 +- *(1) Sort [Time#16 ASC NULLS FIRST, RelativeTimeMilliseconds#2223 ASC NULLS FIRST, Clock#18 ASC NULLS FIRST, HR#2290 ASC NULLS FIRST, ST_II#2357 ASC NULLS FIRST, Pulse#2424 ASC NULLS FIRST, SpO2#2491 ASC NULLS FIRST, Perf#23 ASC NULLS FIRST, etCO2#2625 ASC NULLS FIRST, ImCO2#2692 ASC NULLS FIRST, awRR#2759 ASC NULLS FIRST, NBP_Sys#2826 ASC NULLS FIRST, NBP_Dia#2893 ASC NULLS FIRST, NBP_Mean#2960 ASC NULLS FIRST, NBP_Pulse#3027 ASC NULLS FIRST, NBP_Time_Remaining#3094 ASC NULLS FIRST, ART_Sys#3161 ASC NULLS FIRST, ART_Dia#3228 ASC NULLS FIRST, ART_Mean#3295 ASC NULLS FIRST, etDES#35 ASC NULLS FIRST, inDES#36 ASC NULLS FIRST, etISO#37 ASC NULLS FIRST, inISO#38 ASC NULLS FIRST, etSEV#39 ASC NULLS FIRST, ... 45 more fields], false, 0 +- *(1) Project [Time#16, cast(RelativeTimeMilliseconds#17 as double) AS RelativeTimeMilliseconds#2223, Clock#18, cast(HR#19 as double) AS HR#2290, ST-II#20 AS ST_II#2357, cast(Pulse#21 as double) AS Pulse#2424, cast(SpO2#22 as double) AS Sp...]	Disk Memory Deserialized 1x Replicated	12	100%	0.0 B	376.2 MiB
2306	*(1) Filter ((CrossValidator_10bae398f196_rand#171529 < 0.3333333333333333) OR (CrossValidator_10bae398f196_rand#171529 >= 0.6666666666666666)) +- *(1) Project (Time#16, RelativeTimeMilliseconds#2223, Clock#18, HR#2290, ST_II#2357, Pulse#2424, SpO2#2491, Perf#2558, etCO2#2625, ImCO2#2692, awRR#2759, NBP_Sys#2826, NBP_Dia#2893, NBP_Mean#2960, NBP_Pulse#3027,	Disk Memory Deserialized 1x Replicated	12	100%	891.9 MiB	0.0 B

Executors

[+ Show Additional Metrics](#)

Summary

	* RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Excluded
Active(2)	36	556.4 MiB / 2.8 GiB	4.7 GiB	4	4	0	8639	8643	1.7 h (7.6 min)	228 GiB	75.8 MiB	75.4 MiB	0
Dead(0)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B	0
Total(2)	36	556.4 MiB / 2.8 GiB	4.7 GiB	4	4	0	8639	8643	1.7 h (7.6 min)	228 GiB	75.8 MiB	75.4 MiB	0

Executors

Show 20 entries

Search:

Executor ID	* Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Logs	Thread Dump
0	172.18.0.5:40447	Active	36	556 MiB / 2 GiB	4.7 GiB	4	4	0	8639	8643	1.7 h (7.6 min)	228 GiB	75.8 MiB	75.4 MiB	stdout stderr	Thread Dump
driver	9de5d30478c8:38883	Active	0	382.7 KiB / 912.3 MiB	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B		Thread Dump

COMPLETED:

Completed Applications (1)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
app-20260327140106-0000	DDS_Project3_VitalSigns	4	4.0 GiB		2026/03/27 14:01:00	root	FINISHED	31 min

```
[train] Loading data from: hdfs://namenode:8020/user/root/dataset/*.csv
```

```
[train] Total valid rows: 4,475,701
```

```
+-----+-----+
|label_classification| count|
+-----+-----+
|                    |    0|4465755|
|                    |    1|  9946|
+-----+-----+
```

```
[train] Splitting data: 70% train / 15% val / 15% test
```

```
26/03/27 14:02:23 WARN package: Truncated the string representation of a plan since it was too large. This behavior can be adjusted by setting 'spark.sql.debug.maxToStringFields'.
```

```
[train] Train: 3,133,346 Val: 671,189 Test: 671,166
```

```
[train] Test split saved → hdfs://namenode:8020/user/root/models/test_split
```

COMPLETED:

```
[train] — Random Forest Classifier —————  
[train] AUC=0.9932  Acc=0.9977  F1=0.9965  P=0.9954  R=0.9977  time=368.5s  
[train] Saved → hdfs://namenode:8020/user/root/models/rf_classifier  
  
[train] — GBT Classifier —————  
[train] AUC=0.9928  Acc=0.9979  F1=0.997  P=0.9979  R=0.9979  time=330.2s  
[train] Saved → hdfs://namenode:8020/user/root/models/gbt_classifier  
  
[train] — Random Forest Regressor —————  
[train] RMSE=3.4608  MAE=2.1832  R2=0.9341  time=362.8s  
[train] Saved → hdfs://namenode:8020/user/root/models/rf_regressor  
  
[train] — GBT Regressor —————  
[train] RMSE=2.7131  MAE=1.4563  R2=0.9595  time=390.3s  
[train] Saved → hdfs://namenode:8020/user/root/models/gbt_regressor
```


COMPLETED:

```
[train] — Metrics Summary —  
[  
  {  
    "model": "RF_Classifier",  
    "train_time_s": 368.5,  
    "auc": 0.9932,  
    "accuracy": 0.9977,  
    "f1": 0.9965,  
    "precision": 0.9954,  
    "recall": 0.9977  
  },  
  {  
    "model": "GBT_Classifier",  
    "train_time_s": 330.2,  
    "auc": 0.9928,  
    "accuracy": 0.9979,  
    "f1": 0.997,  
    "precision": 0.9979,  
    "recall": 0.9979  
  },  
  {  
    "model": "RF_Regressor",  
    "train_time_s": 362.8,  
    "rmse": 3.4608,  
    "mae": 2.1832,  
    "r2": 0.9341  
  },  
  {  
    "model": "GBT_Regressor",  
    "train_time_s": 390.3,  
    "rmse": 2.7131,  
    "mae": 1.4563,  
    "r2": 0.9595  
  }  
]  
[train] Done.
```

/user/root/models Go!   

Show 25 entries Search:

☐	 Permission	 Owner	 Group	 Size	 Last Modified	 Replication	 Block Size	 Name	
☐	drwxr-xr-x	root	supergroup	0 B	Mar 27 15:19	0	0 B	gbt_classifier	
☐	drwxr-xr-x	root	supergroup	0 B	Mar 27 15:32	0	0 B	gbt_regressor	
☐	drwxr-xr-x	root	supergroup	0 B	Mar 27 15:13	0	0 B	rf_classifier	
☐	drwxr-xr-x	root	supergroup	0 B	Mar 27 15:25	0	0 B	rf_regressor	
☐	drwxr-xr-x	root	supergroup	0 B	Mar 27 15:06	0	0 B	test_split	

Showing 1 to 5 of 5 entries Previous 1 Next

Step 6

- Test, run the command:
`docker exec -it spark-master2 /spark/bin/spark-submit --master spark://spark-master:7070 --driver-memory 2G --executor-memory 4G /app/predict.py`
- Test contains Temporal segments – early/mid/late stage of the procedure, and Alarm state – active/no alarms

```
[predict] Test rows: 671,166  
[predict] Loading models ...
```



```
[predict] — Classification Experiments —————
[RF_Classifier][full_test]  AUC=0.9933  F1=0.9966  Acc=0.9977  n=671,166
[GBT_Classifier][full_test] AUC=0.9926  F1=0.997  Acc=0.9979  n=671,166
[RF_Classifier][temporal_early] AUC=0.99  F1=0.9946  Acc=0.9964  n=419,419
[GBT_Classifier][temporal_early] AUC=0.9896  F1=0.9952  Acc=0.9966  n=419,419
[RF_Classifier][temporal_mid]  AUC=0.0  F1=1.0  Acc=1.0  n=173,273
[GBT_Classifier][temporal_mid] AUC=0.0  F1=1.0  Acc=1.0  n=173,273
[RF_Classifier][temporal_late] AUC=0.0  F1=1.0  Acc=1.0  n=78,473
[GBT_Classifier][temporal_late] AUC=0.0  F1=1.0  Acc=1.0  n=78,473
[RF_Classifier][alarms_active] AUC=1.0  F1=0.998  Acc=0.9986  n=101,910
[GBT_Classifier][alarms_active] AUC=0.9983  F1=0.998  Acc=0.9986  n=101,910
[RF_Classifier][no_alarms]  AUC=0.9917  F1=0.9964  Acc=0.9976  n=569,256
[GBT_Classifier][no_alarms]  AUC=0.9918  F1=0.9968  Acc=0.9978  n=569,256
[predict] Metrics saved → hdfs://namenode:8020/user/root/results/cls_metrics

[predict] — Regression Experiments —————
[RF_Regressor][full_test]  RMSE=3.4486  MAE=2.1758  R2=0.9345  n=671,166
[GBT_Regressor][full_test] RMSE=2.7008  MAE=1.4505  R2=0.9598  n=671,166
[RF_Regressor][temporal_early] RMSE=3.9301  MAE=2.5015  R2=0.9216  n=419,419
[GBT_Regressor][temporal_early] RMSE=3.1752  MAE=1.7126  R2=0.9488  n=419,419
[RF_Regressor][temporal_mid]  RMSE=2.6593  MAE=1.7347  R2=0.9606  n=173,273
[GBT_Regressor][temporal_mid] RMSE=1.8192  MAE=1.0939  R2=0.9816  n=173,273
[RF_Regressor][temporal_late] RMSE=1.8833  MAE=1.4089  R2=0.7619  n=78,473
[GBT_Regressor][temporal_late] RMSE=1.0936  MAE=0.8368  R2=0.9197  n=78,473
[RF_Regressor][alarms_active] RMSE=3.8065  MAE=2.4928  R2=0.9251  n=101,910
[GBT_Regressor][alarms_active] RMSE=2.2358  MAE=1.4504  R2=0.9742  n=101,910
[RF_Regressor][no_alarms]  RMSE=3.3805  MAE=2.119  R2=0.9364  n=569,256
[GBT_Regressor][no_alarms]  RMSE=2.7758  MAE=1.4505  R2=0.9571  n=569,256
[predict] Metrics saved → hdfs://namenode:8020/user/root/results/reg_metrics
```

```
[predict] Total inference time: 257.3s
```

[predict] — Final Summary

Classification:

RF_Classifier	[full_test	] AUC=0.9933 F1=0.9966
GBT_Classifier	[full_test	] AUC=0.9926 F1=0.997
RF_Classifier	[temporal_early	] AUC=0.99 F1=0.9946
GBT_Classifier	[temporal_early	] AUC=0.9896 F1=0.9952
RF_Classifier	[temporal_mid	] AUC=0.0 F1=1.0
GBT_Classifier	[temporal_mid	] AUC=0.0 F1=1.0
RF_Classifier	[temporal_late	] AUC=0.0 F1=1.0
GBT_Classifier	[temporal_late	] AUC=0.0 F1=1.0
RF_Classifier	[alarms_active	] AUC=1.0 F1=0.998
GBT_Classifier	[alarms_active	] AUC=0.9983 F1=0.998
RF_Classifier	[no_alarms	] AUC=0.9917 F1=0.9964
GBT_Classifier	[no_alarms	] AUC=0.9918 F1=0.9968

Regression:

RF_Regressor	[full_test	] RMSE=3.4486 $R^2=0.9345$
GBT_Regressor	[full_test	] RMSE=2.7008 $R^2=0.9598$
RF_Regressor	[temporal_early	] RMSE=3.9301 $R^2=0.9216$
GBT_Regressor	[temporal_early	] RMSE=3.1752 $R^2=0.9488$
RF_Regressor	[temporal_mid	] RMSE=2.6593 $R^2=0.9606$
GBT_Regressor	[temporal_mid	] RMSE=1.8192 $R^2=0.9816$
RF_Regressor	[temporal_late	] RMSE=1.8833 $R^2=0.7619$
GBT_Regressor	[temporal_late	] RMSE=1.0936 $R^2=0.9197$
RF_Regressor	[alarms_active	] RMSE=3.8065 $R^2=0.9251$
GBT_Regressor	[alarms_active	] RMSE=2.2358 $R^2=0.9742$
RF_Regressor	[no_alarms	] RMSE=3.3805 $R^2=0.9364$
GBT_Regressor	[no_alarms	] RMSE=2.7758 $R^2=0.9571$

[predict] Done.

/user/root/results

Go

Show

25

entries

Search

☐	 Permission	 Owner	 Group	 Size	 Last Modified	 Replication	 Block Size	 Name	
☐	drwxr-xr-x	root	supergroup	0 B	Mar 27 21:17	0	0 B	cls_metrics	
☐	drwxr-xr-x	root	supergroup	0 B	Mar 27 20:48	0	0 B	cls_predictions	
☐	drwxr-xr-x	root	supergroup	0 B	Mar 27 21:18	0	0 B	reg_metrics	
☐	drwxr-xr-x	root	supergroup	0 B	Mar 27 15:32	0	0 B	training_metrics	

Showing 1 to 4 of 4 entries

Previous

1

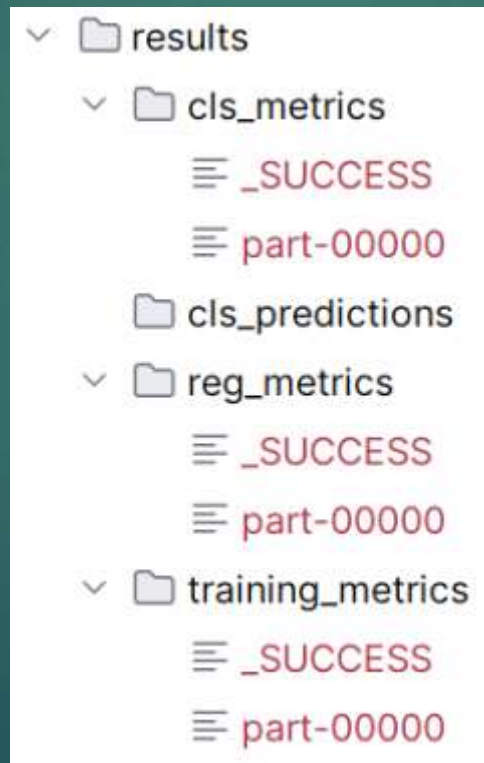
Next

Step 7

- Pulling the results, run:

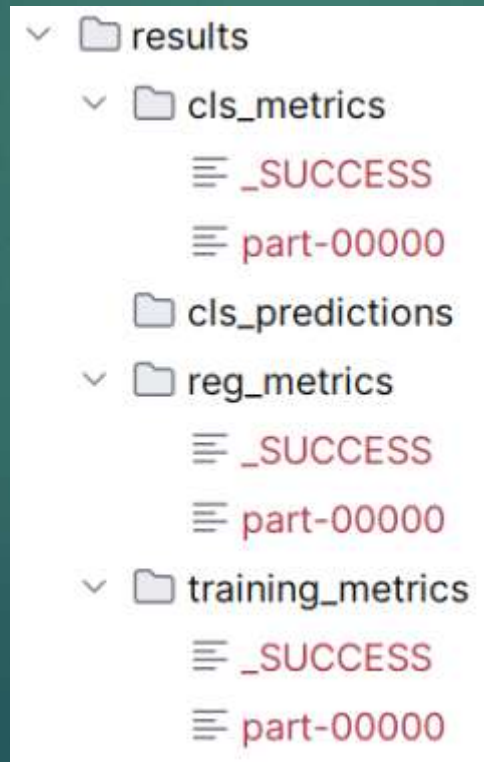
```
docker exec namenode hdfs dfs -get /user/root/results  
/data/incoming/
```

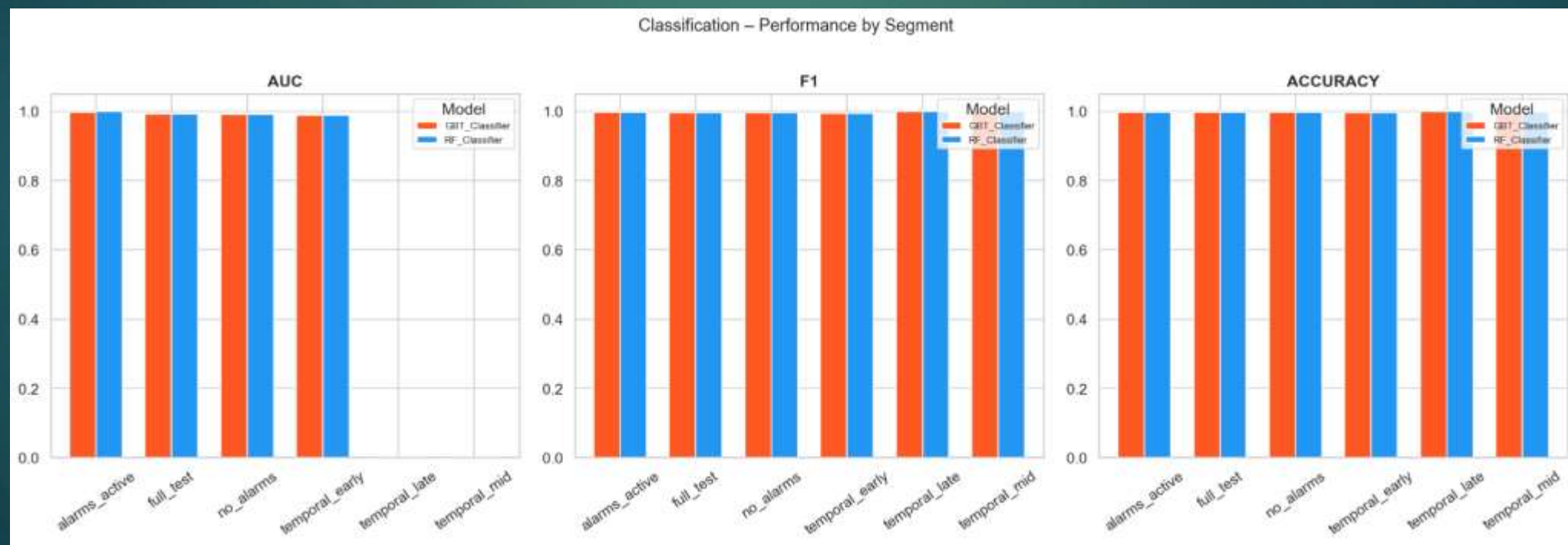
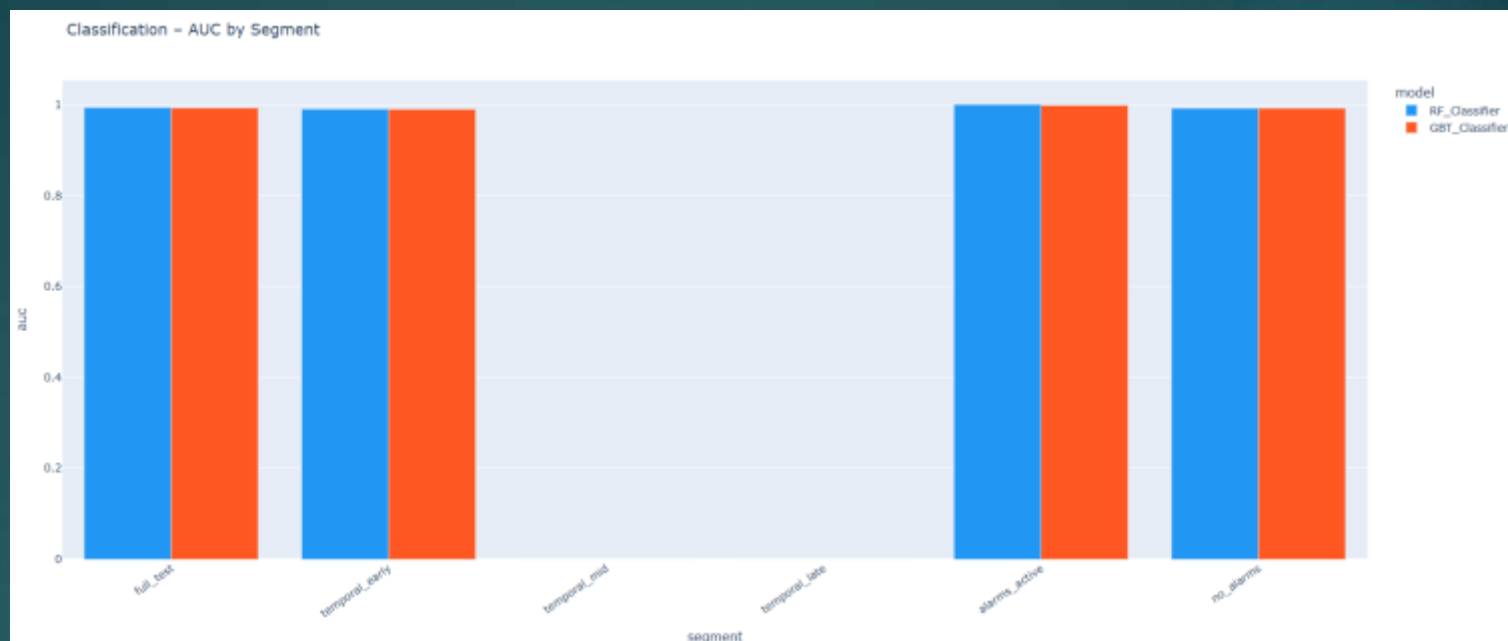
```
docker cp namenode:/data/incoming/results ./results
```

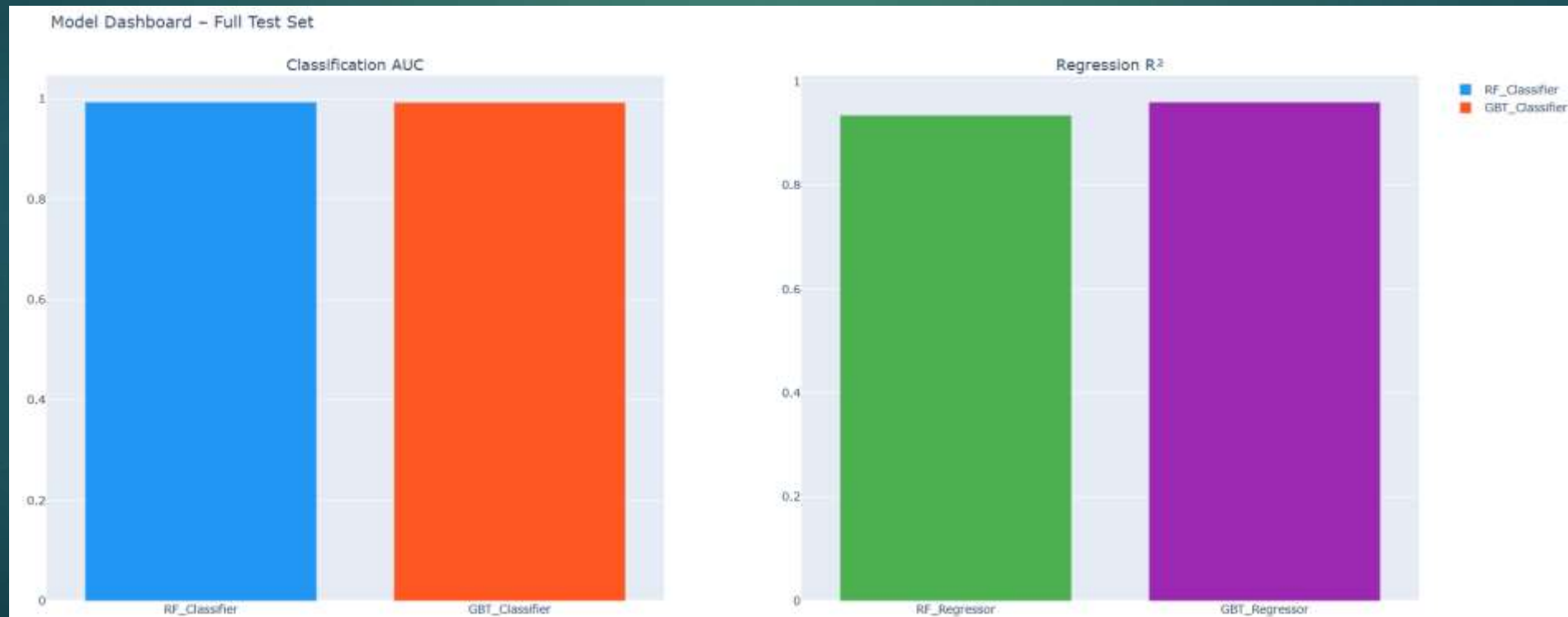
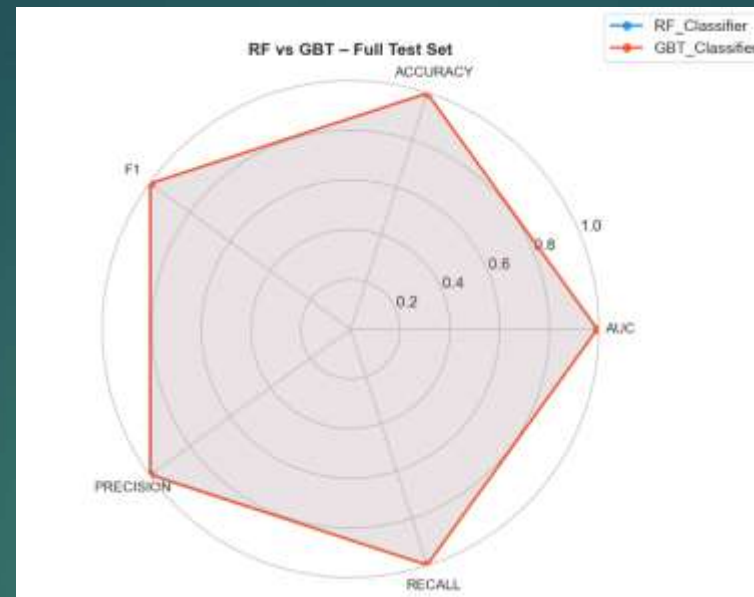


Step 8

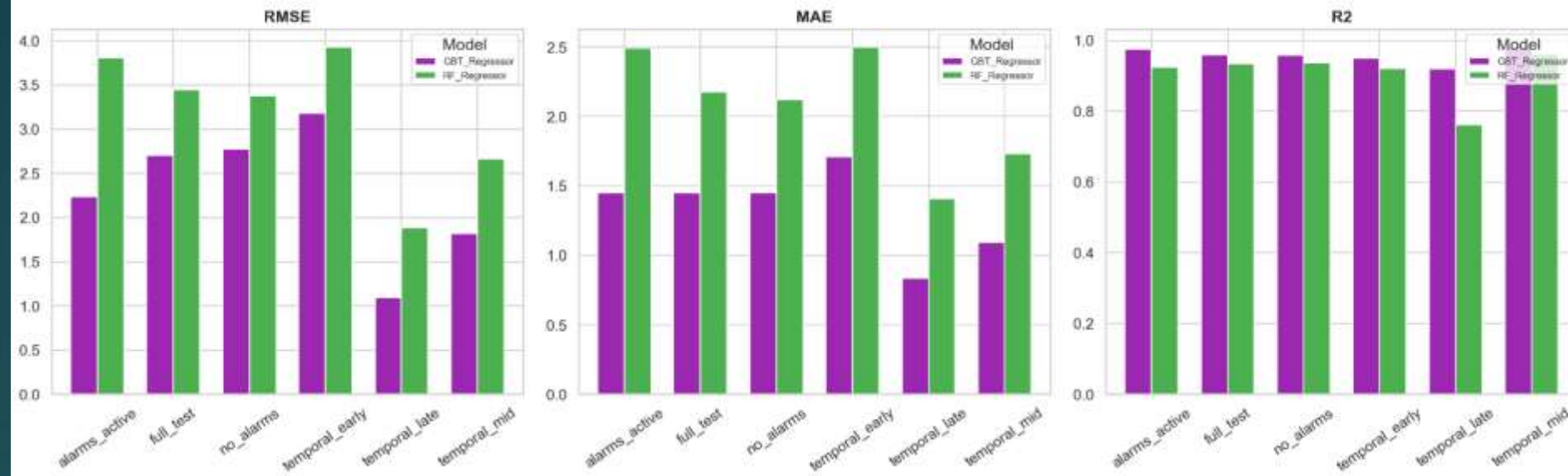
- Visualization, run:
`python visualize.py --results ./results --output ./results/plots`







Regression (HR) – Performance by Segment



Regression – RMSE vs MAE

